PREGNANCY SUPPORT BELT
4-Month Development Roadmap

A practical engineering schedule from requirements and component selection to an integrated, tested and documented prototype.

Month 1 Define & select	**Month 2** Build electronics	**Month 3** Integrate	**Month 4** Verify & document
Requirements, risk register, block diagram, preliminary BOM and procurement.	ESP32 + GY-521 tests, firmware, power measurements, KiCad schematic draft.	Calibration, battery integration, belt mock-up, data logging.	Repeatability/runtime tests, final BOM, report and demo.

Planning assumptions & safety boundary

Assumes an ESP32 development board and GY-521 breakout (commonly based on MPU-6050) for motion/orientation experiments. Optional BLE, LED or haptic functions should be included only if confirmed in requirements. This is an engineering prototype plan, not a validated medical-device plan; do not use prototype readings for pregnancy-care decisions. Any clinical claims or human-subject testing require appropriate expert oversight and relevant ethics/regulatory review.

Dates are relative to project kickoff. Adjust for procurement delays and academic milestones. Check prices and availability when ordering.

1. Month-by-month work plan

Period	Tasks	Deliverables	Exit criteria
Month 1 Requirements & selection	Define intended use, sensing features, constraints and success criteria. Compare controller, IMU, battery, charger, regulator, switch, connectors and enclosure. Check datasheets, lifecycle, availability and lead times. Draft block diagram, risk register, budget and test plan.	Requirements sheet; component selection matrix; preliminary BOM with links; block diagram; procurement tracker; risk list.	Scope agreed; key parts sourceable; limitations and safety boundaries documented.
Month 2 Electronics & firmware	Bring up ESP32 over USB. Connect GY-521 via I²C and log accelerometer/gyroscope data. Verify supply rails and restart behaviour. Measure current. Draft KiCad schematic, connector definitions and footprints; review electrical rules.	Working firmware; sensor logs; wiring diagram; schematic draft; power measurements and issue log.	Repeatable sensor communication and stable bench operation; issues tracked with fixes.
Month 3 Integration & mock-up	Calibrate sensor and define processing approach. Add optional BLE/button/LED only if required. After measuring current, integrate protected battery/charging. Build removable electronics pod and textile mounting with strain relief. Review fit using a non-powered mock-up first.	Integrated prototype; calibration notes; sample logs; updated BOM/schematic; mechanical fit review.	Electronics removable for cleaning; battery secured; wiring strain-relieved; sensor placement repeatable.
Month 4 Verification & handover	Test sensor repeatability, runtime, charging, restart and disconnection behaviour. Review drift and data quality. Resolve high-priority issues; freeze component choices. Complete final KiCad files, BOM, assembly notes, test report and demo.	Verification matrix; final BOM and sourcing notes; schematic; prototype documentation; presentation and future-work list.	Every agreed engineering requirement has a recorded result or documented limitation; open risks are explicit.

2. Weekly checkpoints & engineering targets

Weeks	Checkpoint	Evidence to retain
1–2	Scope, requirements, risks and acceptance criteria defined.	Requirements document, risk register, block diagram.
3–4	Component shortlist approved; long-lead items ordered.	BOM with links, datasheets, procurement tracker.
5–6	ESP32 toolchain works; IMU detected over I²C.	Firmware, serial logs, wiring diagram.
7–8	Sensor readings stable; current measured; schematic reviewed.	Sensor logs, power measurements, KiCad review notes.
9–10	Calibration and data logging work; optional functions only if needed.	Calibration method, sample data, software version.
11–12	Removable electronics pod and belt mock-up reviewed.	Fit review, photos, mechanical revisions.
13–14	Power, charging and runtime tests completed.	Test records, battery/charger details, issue log.
15–16	Verification, final documentation and demo completed.	Test matrix, final BOM, schematic, presentation.

Preliminary engineering targets (confirm after requirements are fixed)